

Gelos: Incident Response Exercise (IRX)

Implementation Submission

Submission for: Part 5: Implement the Incident Response Exercise (IRX) Project Plan

Team Name: Delta Soc

Date: 16/06/2026

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1. Updated Project Plan Document

Gelos: Incident Response Exercise (IRx) Project Plan

Version: 2 Date: 07/06/2026 Prepared by: Delta Soc, DataTrust

Team Charter / Agreement

Project Name

Gelos Enterprise's Incident Response Exercise (IR) Project

Team Members

- **[Andrew Byrne]** (Team Lead)
- [Team member] (Scenario Designer)
- [Team member] (Environment Admin)
- [Team member] (Observer/Evaluator)
- [Team member] (Observer/Evaluator)

Roles & Responsibilities

Each team member agrees to complete the tasks assigned to their role for this project in accordance with the schedule, including development of their assigned IRx project section and corresponding Red Team playbook, and to participate in all team meetings and exercises.

Communication

Communication will be conducted via Microsoft Teams, with documents shared through the associated SharePoint workspace. Weekly team meetings will be held every Wednesday and formally documented to establish and review the weekly work plan. Ongoing communication will be maintained through the Teams chat and email.

Conflict Resolution

All conflicts will be managed professionally through open discussion during scheduled team meetings, with the aim of reaching a resolution through consensus.

Where necessary, decisions will be determined through structured group discussion or majority agreement. If a resolution cannot be achieved at the team level, the issue will be escalated through appropriate corporate channels in line with organisational governance processes.

Commitment

Each team member commits to providing an equal and fair contribution to all assigned project deliverables, meeting agreed deadlines, and actively participating in all meetings and project activities.

Signatures of all team members:

- [Student Name 1]: Team member
- [Student Name 2]: Andrew Byrne
- [Student Name 3]: Team member
- [Student Name 4]: Team member
- [Student Name 5]: Team member

Individual Contribution Log / Statement

Andrew Byrne (Team Lead)

Student ID: Not included in public portfolio version

As Team Lead, I was responsible for coordinating the development of the Incident Response Exercise (IRx) Project Plan. My contributions included organising and leading team meetings, defining the overall document structure, and assigning tasks to ensure balanced workload across the team.

I developed key sections of the project plan including the Team Charter, communication strategy, and project objectives, ensuring alignment with previous research and the Incident Response Plan developed in Part 3. I also reviewed and consolidated team contributions to maintain consistency in writing style, structure, and overall quality.

In addition, I developed the Red Team playbook for Privilege Escalation, aligning it with the Blue Team playbook created in Part 3 and ensuring it could be effectively executed during the IRx. Detailed Contributions are listed in part 5 of the AAR.

Team member (Scenario Designer)

Student ID: Not included in public portfolio version

As a Scenario Designer, I conducted external research and gathered information on relevant industry standards, guidelines, codes of practice, and best practices applicable to the identified cyber security issues and incident response. As part of Gelos's Current Cyber Security Posture Analysis, I outlined Gelos's existing security clearance levels related to organisational data, as per their policies, and provided an assessment of the valuation of organisational assets (e.g., specific data types, critical systems, intellectual property) that are particularly impacted by or require protection under Gelos's cyber security policies.

As part of my involvement in developing our Incident Response Plan (IRP), I have recruited and defined IRT Members i.e. the proposed members of Gelos's Incident Response Team. And for each IRT member:

- I have defined their specific roles and responsibilities within the IRT (e.g., Incident Commander, Forensic Analyst, Communications Lead, Legal Counsel representative).
- I have outlined the essential training requirements necessary for each role to ensure they can effectively perform their incident response duties.

I assisted in developing the overall structure of the IRx, including ensuring that the planned attack scenarios were realistic, relevant, and achievable within the lab environment.

In addition, I developed the Red Team playbook for the IDOR (Insecure Direct Object Reference) attack and ensuring that the attack steps, expected behaviour, and outcomes in that Red Team Playbook are aligned with the corresponding Blue Team playbook that I also developed previously.

Detailed Contributions are listed in part 5 of the AAR.

Team member (Environment Admin)

Student ID: Not included in public portfolio version

As Environment Administrator, I contributed to the planning of the IRx by outlining the technical requirements and structure of the lab environment. My work included supporting the definition of system components, resources, and configurations required to successfully execute the simulated attacks and observe defensive responses.

I was responsible for developing the implementation plan, including the Statement of Work (SOW), Work Breakdown Structure (WBS), and project milestones, ensuring that the project timeline was realistic and achievable.

In addition, I developed the Red Team playbook for the Ransomware attack, ensuring the steps were clearly defined and aligned with the technical capabilities of the Healthcare VM environment.

Detailed Contributions are listed in part 5 of the AAR.

Team member (Observer/Evaluator)

Student ID: Not included in public portfolio version

As Observer/Evaluator, I contributed to defining the project scope, system boundaries, and problem-solving methodology for the IRx. My work focused on ensuring that the exercise operated within a clearly defined and controlled lab environment and that all activities were aligned with the project objectives.

I also contributed to designing how the exercise would be evaluated, including identifying key behaviours, actions, and outcomes that will be observed during the IRx.

In addition, I developed the Red Team playbook for the reconnaissance phase of the attack chain, ensuring it generates measurable indicators that can be detected and analysed by the Blue Team during the exercise.

Detailed Contributions are listed in part 5 of the AAR.

Team member (Observer/Evaluator)

Student ID: Not included in public portfolio version

As Observer/Evaluator, I contributed to the development of the IRx Project Plan by defining key project deliverables, expected outcomes, and team performance evaluation criteria. My

work focused on ensuring that the results of the exercise can be measured and used to assess the effectiveness of the Incident Response Plan.

I also developed the risk management plan, including identifying potential risks to the successful execution of the IRx and defining mitigation strategies and required resources.

In addition, I developed the Red Team playbook for the SQL Injection attack, ensuring that the attack process, expected results, and generated indicators align with the Blue Team's detection and response capabilities.

Detailed Contributions are listed in part 5 of the AAR.

1. Project Background & Problem Statement

Cyber security threats are rapidly increasing, which results in an increased risk of online crime, information theft, and exploitation for all individuals and organisations. Gelos Enterprises is no exception to these cyber security threats. They suspected that a data breach may have occurred through their outdated publicly accessible system / service along with employees' habits that were identified in our Task 2 research, which may have allowed attackers to commit identity theft by stealing sensitive customer information of up to 3.4 million customers such as home addresses and government IDs (10,000 passport numbers and 15,000 driver's licence numbers). This can have long-term financial and credit score consequences for affected individuals.

Now that Gelos has a new Incident Response Plan (IRP) along with Blue Team playbooks (developed in Task 3), it is important to make sure through a simulated Incident Response Exercise (IRX), that these plans and blue playbooks can work in a real-world workplace scenario. This will test whether or not the IRP and the Blue Team are prepared enough for any cyber security threat that they can face. Therefore, Gelos will be ready to respond to real-world cyber incidents which reduces their financial and reputational damage.

2. Project Scope, System Boundaries, and Problem-Solving Methodology

Project Scope

This project will plan, configure, execute, and evaluate a simulated Incident Response Exercise (IRX) designed to test the effectiveness of Gelos Enterprises' Incident Response Plan (IRP) and the Blue Team playbooks developed in Task 3. The exercise will focus specifically on validating incident response capabilities across five common cyber attack scenarios: Reconnaissance (Recon), SQL Injection (SQLi), Insecure Direct Object Reference (IDOR),

Reverse Shell activity, and Privilege Escalation. The project will assess whether the organisation's response processes can operate effectively under realistic attack conditions and support rapid containment and recovery.

Specific activities included within scope are scenario design and development, configuration of the testing environment, deployment of security monitoring and logging tools, execution of controlled attack simulations, Blue Team response activities, evidence collection, and post-exercise evaluation. Each scenario will test different phases of the IRP including detection, analysis, containment, eradication, recovery, and post-incident review to identify strengths and improvement opportunities within current response procedures.

System Boundaries

The Incident Response Exercise will be conducted within a dedicated laboratory environment designed to simulate relevant components of Gelos Enterprises' infrastructure. The exercise environment will remain fully isolated from production systems to ensure business operations and customer data are not affected during testing activities. Simulated environments and synthetic data will be used instead of live operational assets.

The exercise will include configuration and monitoring of security infrastructure components necessary to support the selected attack scenarios. These components include firewalls for traffic filtering and access control, Intrusion Detection Systems (IDS) for identifying suspicious activity, Security Information and Event Management (SIEM) platforms for alerting and log analysis, endpoint monitoring controls, system logging, and segmented network environments. The scope of testing will include systems and controls required to observe reconnaissance behavior, detect SQL injection attempts, monitor unauthorised object access, identify reverse shell communications, and detect privilege escalation attempts. Limitations include time constraints, controlled attack execution, and the inability to replicate all variables and unpredictability present in real-world cyber incidents.

Problem-Solving Methodology

This project will use a phased methodology to manage the planning and execution of the Incident Response Exercise. The phases will consist of preparation and planning, environment setup and configuration, attack simulation execution, incident response observation, and post-exercise evaluation. This methodology was selected because testing multiple attack scenarios requires a structured and sequential approach that allows each stage to be completed, validated, and reviewed before progressing.

A phased approach is suitable for this project because it supports controlled testing while reducing operational complexity across multiple cyber attack scenarios. Separating the exercise into stages ensures that Recon, SQL Injection, IDOR, Reverse Shell, and Privilege Escalation incidents can be individually assessed while maintaining consistency in evaluation criteria. This methodology also improves the quality of findings by enabling systematic analysis of Blue Team performance, identifying gaps in the IRP and playbooks, and producing actionable recommendations to strengthen Gelos Enterprises' cyber security readiness.

3. Project Objectives and Key Questions

Project Objectives

- Test all blue playbooks that were developed in Task 3 to determine whether these are effective in a simulated real-world cyber attack.
- Assess and evaluate Gelos' current SIEM and IDS configurations and the Blue Team's capability to identify simulated real-world cyber attacks
- Analyse and detect any areas for improvement of Gelos' IRP after the simulated Incident Response Exercise (IRX) was conducted.

Key Questions the Project Will Answer

- What challenges and issues can the Incident Response Team can expect face in a real-world work environment when executing their IRP?
- Will the Incident Response Team know what steps to take from their IRP based on the cyber attack they face?
- Will the Blue Team be able to quickly and accurately identify any cyber attack during the IRX using the tools that they have?
- Does the blue playbook provide step-by-step log collection instructions for their specific threat?

Key Project Deliverables and Expected Outcomes

4. Key Project Deliverables and Expected Outcomes

Deliverable	Expected Outcome	Status
Red Team Activity Playbooks	Provides structured attack scenario guides that can be safely executed in the lab environment to test Gelos's Incident Response Plan and validate the Blue Team playbooks developed in Part 3. Each team member will develop one Red Team activity/attack playbook as required by the assessment.	Completed
Configured Exercise Environment	Provides a dedicated, ready-to-use lab network environment that mimics Gelos's production environment closely enough for exercise purposes. The environment will include configured firewall, IDS, and SIEM components so Red Team attacks and Blue Team monitoring can be tested safely without disrupting Gelos's real operations.	Completed
Exercise Plan B — Contingency Scenario	Provides a backup method for continuing the IRx if live attack execution fails due to technical problems. The contingency plan will use an injected PCAP file to simulate a specific incident, allowing the Blue Team to continue detection, analysis, and response activities.	Completed
Purple Teaming Activities Plan and Observation Checklist	Provides a structured approach for Red Team and Blue Team collaboration during the exercise. The observation checklist will be used to monitor team actions, detection quality, response timing, communication, evidence capture, and lessons learned.	Completed
IRx After-Action Report (AAR)	Provides a formal evaluation of the effectiveness of Gelos's Incident Response Plan based on the IRx results. The AAR will document exercise objectives, findings, IRP strengths and weaknesses, recommendations, and conclusions.	Completed
Exercise Evidence Pack	Provides labelled screenshots, logs, notes, checklist entries, and other supporting evidence captured during the IRx. This evidence will support the findings in the After-Action Report and demonstrate that the exercise was completed as planned. Part 5 requires compiled evidence from IRx execution to be clearly labelled and contextualised.	Completed

Project Management Documentation	Provides evidence that the project was managed professionally, including task allocation, actual start/end dates, deviations from the original plan, meeting minutes, decision logs, and risk management updates. Part 5 requires the updated project plan and project management documentation.	Completed

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Implementation / Work Plan Schedule (Updated with Actual)

Statement of Work (SOW):

DataTrust designed, configured, executed, and evaluated a simulated Incident Response Exercise (IRX) for Gelos Enterprises. The exercise tested five attack scenarios against the Blue Team's Incident Response Plan (IRP): Ransomware (Team member), IDOR (Team member), SQL Injection (Team member), Privilege Escalation (Andrew Byrne), and Network Reconnaissance (Team member). Each team member developed and executed their assigned Red Team attack, while the Blue Team monitored, detected, and responded using the IRP developed in Part 3.

Task ID	Task Description	Planned Start	Planned End	Actual Start	Actual End	Notes on Deviations/Execution
1.0	Planning Phase	22/04/2026	06/05/2026	22/04/2026	13/05/2026	Slight delay Red Team Playbooks completed one week late due to Part 3 overlap
1.1	Define Exercise Scenarios & Objectives	22/04/2026	29/04/2026	22/04/2026	06/05/2026	1 week delay finalised at Week 12 meeting (06/05)
1.2	Develop Red Team Playbooks	29/04/2026	06/05/2026	06/05/2026	13/05/2026	1 week delay, team members completing Part 3 concurrently
1.3	Exercise Plan B (PCAP Contingency)	29/04/2026	06/05/2026	29/04/2026	06/05/2026	On schedule
1.4	Purple Team Activities & Observation Checklist	29/04/2026	06/05/2026	29/04/2026	06/05/2026	On schedule
2.0	Environment Setup Phase	07/05/2026	20/05/2026	07/05/2026	20/05/2026	On schedule overall
2.1	Configure Lab Network (VMs)	07/05/2026	13/05/2026	07/05/2026	20/05/2026	1 week delay lab configuration extended to align with IRX week
2.2	Configure Firewall, IDS, SIEM	13/05/2026	20/05/2026	13/05/2026	20/05/2026	On schedule. Zeek interface misconfiguration noted during exercise.
3.0	IRX Execution Phase	21/05/2026	27/05/2026	20/05/2026	27/05/2026	On schedule. Team member's attack

Task ID	Task Description	Planned Start	Planned End	Actual Start	Actual End	Notes on Deviations/Execution
						executed 20/05; full team IRX 26–27/05.
3.1	Red Team Attacks Round 1 (All Members)	21/05/2026	23/05/2026	20/05/2026	27/05/2026	Ransomware Team member (20/05); IDOR Team member (26–27/05); SQL Injection Team member; Privilege Escalation Andrew Byrne; Network Reconnaissance Team member. All attacks executed within the exercise window.
3.2	Blue Team Response & Observation (All Members)	21/05/2026	23/05/2026	26/05/2026	27/05/2026	Blue Team monitored and responded to all 5 attacks. Conducted 26–27/05 alongside Round 2.
3.3	Role Swap & Second Round	24/05/2026	27/05/2026	26/05/2026	27/05/2026	On schedule
4.0	Evaluation & Reporting Phase	28/05/2026	19/06/2026	02/06/2026	19/06/2026	Started slightly late post-IRX debrief held at Week 16 meeting 02/06. On track for 19/06/2026.
4.1	Analyse Exercise Results	28/05/2026	02/06/2026	02/06/2026	02/06/2026	Discussed at Week 16 meeting 02/06
4.2	Draft IRX After-Action Report (AAR)	02/06/2026	19/06/2026	02/06/2026	19/06/2026	In progress on track for 19/06/2026
4.3	Part 5 Documentation & Submission	02/06/2026	19/06/2026	02/06/2026	19/06/2026	In progress on track for 19/06/2026

Total Estimated Project Cost:

\$3,500 (Actual costs: \$3,500, On budget. All work was completed within the lab environment using pre-allocated tools and resources. No unplanned expenditure was incurred.)

Key Milestones Met:

Milestone 1: Red Team Playbooks Completed Planned: 06/05/2026 | Actual: 13/05/2026
One week delay due to team members finalising Part 3 concurrently.

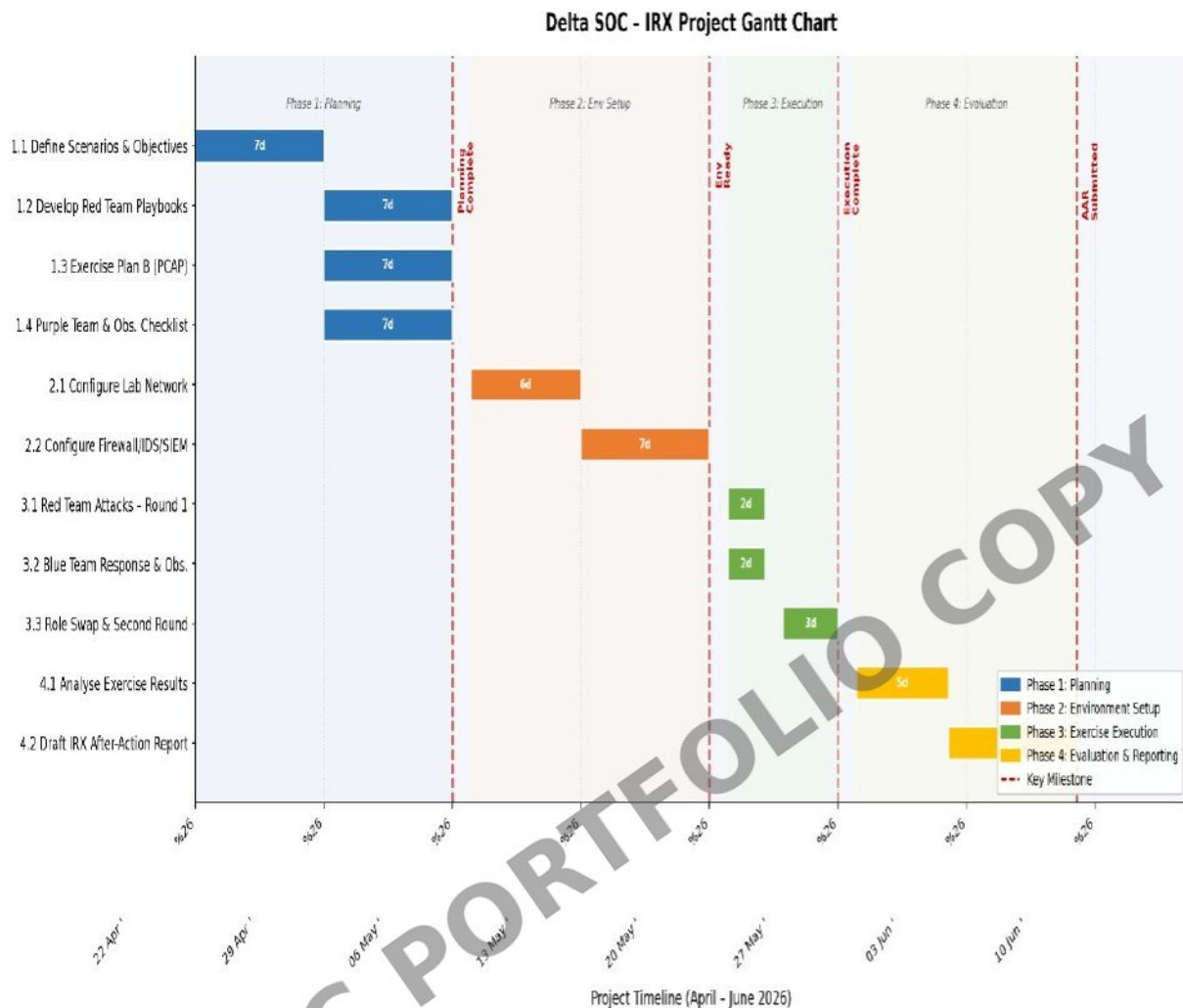
Milestone 2: Lab Environment Ready — Planned: 20/05/2026 | Actual: 20/05/2026, On schedule.

Milestone 3: IRX Exercise Completed — Planned: 27/05/2026 | Actual: 27/05/2026, On schedule. Ransomware attack executed 20/05/2026; full team IRX conducted 26–27/05/2026.

Milestone 4: AAR & Part 5 Documentation Complete Planned: 19/06/2026 | Actual: In progress, On track for submission by 19/06/2026.

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Gantt Chart



A Gantt chart has been developed based on the Work Breakdown Structure above. The chart maps all four project phases across the project timeline from 22/04/2026 to 09/06/2026, showing task durations, dependencies, and key milestones. The chart is maintained in the team's SharePoint workspace under the Delta SOC project folder.

Project Team Structure and Performance

Project Team Members and Roles

The Incident Response Exercise (IRX) project team is structured to ensure that all aspects of planning, execution, and evaluation are effectively managed. Each team member has been assigned a primary role aligned with their area of responsibility and their allocated Red Team playbook.

The project team consists of the following members:

- **Team Lead (Andrew Byrne)**
- Overall coordination and project oversight
- Responsible for the **Privilege Escalation Red Team playbook**
- Scenario Designer (Team member)
- Scenario development and alignment with IR objectives
- Responsible for the **IDOR (Insecure Direct Object Reference) Red Team playbook**
- Environment Administrator (Team member)
- Lab setup and technical environment configuration
- Responsible for the **Ransomware Red Team playbook**
- Observer/Evaluator (Team member)
- Exercise monitoring and performance observation
- Responsible for the **Reconnaissance Red Team playbook**
- Observer/Evaluator (Team member)
- Evaluation criteria and outcome analysis
- Responsible for the **SQL Injection Red Team playbook**

This structure ensures that each team member is accountable for both their functional role within the IRX and the development of a corresponding attack scenario.

Team Member Roles and Responsibilities (Specific to IRX Tasks)

To support execution of the Incident Response Exercise (IRX), all team members require access to a shared set of core resources and tools, along with additional role-specific resources aligned with their assigned responsibilities and Red Team playbooks.

Shared Resources (All Team Members)

All team members require access to the following:

- **IRX Lab Environment**
- Isolated attack network environment used for all simulated activities
- **Target Systems**
- Healthcare VM (primary target for reconnaissance, SQL injection, and IDOR scenarios)
- UbuntuX machines (used for ransomware and privilege escalation scenarios)
- **SOC Monitoring Systems**
- Security Onion SIEM for log aggregation, monitoring, and alerting
- **Collaboration & Documentation Tools**
- Microsoft Teams (communication)
- SharePoint (document storage and version control)

Team Lead (Andrew Byrne – Privilege Escalation Playbook)

Tasks:

- Coordinate all project activities and ensure alignment with objectives
- Manage timelines, milestones, and communication
- Review and consolidate deliverables
- Develop and execute the **Privilege Escalation attack playbook**
- Ensure alignment with the corresponding Blue Team playbook

Role-Specific Resources:

- **Copyfail Privilege Escalation PoC**
- Access to UbuntuX target system
- Process monitoring and system logging tools within Security Onion

Scenario Designer (Team member – IDOR Playbook)

Tasks:

- Design realistic and relevant attack scenarios
- Develop the **IDOR playbook**
- Define expected attack behaviour and outcomes
- Ensure alignment with IRP scenarios

Role-Specific Resources:

- Access to Healthcare VM web application
- Web browser developer tools and HTTP manipulation tools (e.g. Burp Suite or similar)
- Application logs via Security Onion

Environment Administrator (Team member – Ransomware Playbook)

Tasks:

- Configure and maintain the lab environment
- Ensure systems, SIEM, and network components are operational
- Develop and execute the **Ransomware playbook**
- Support safe execution of attack simulations

Role-Specific Resources:

- **SWAKs tool (for phishing email simulation)**
- Access to UbuntuX target systems
- File system monitoring and SIEM logging tools
- Backup and recovery systems for controlled restoration

Observer/Evaluator (Team member – Reconnaissance Playbook)

Tasks:

- Monitor reconnaissance activity and early-stage attack behaviour
- Record indicators of compromise and detection points
- Develop the **Reconnaissance playbook**
- Evaluate early detection capability

Role-Specific Resources:

- Network scanning tools (e.g. Nmap or equivalent)
- Network traffic visibility via Security Onion
- Logging and alert data for analysis

Observer/Evaluator (Team member – SQL Injection Playbook)

Tasks:

- Define performance evaluation criteria and success metrics
- Develop the **SQL Injection playbook**
- Monitor and evaluate detection of application-layer attacks
- Contribute to exercise outcome analysis

Role-Specific Resources:

- Access to Healthcare VM web application
- SQL injection testing tools (e.g. SQLMap or equivalent)
- Database and web server logs via Security Onion

Criteria for Evaluating Team Performance

Team performance during the Incident Response Exercise (IRX) will be evaluated using defined criteria to ensure that all tasks are completed effectively, professionally, and in alignment with project objectives.

The following criteria will be applied:

- **Timeliness of Deliverables**
 - Completion of assigned tasks and milestones within agreed timeframes
 - Adherence to the schedule defined in the Work Breakdown Structure (WBS)
- **Quality and Accuracy of Work**
 - Technical correctness and completeness of playbooks and documentation
 - Alignment of outputs with the Incident Response Plan (Part 3)
 - Clarity, structure, and professionalism of written deliverables
- **Execution of Assigned IRX Tasks**
 - Successful development and execution of assigned Red Team playbooks
 - Ability to produce expected attack behaviours and test scenarios effectively

- Proper use of tools and techniques within the lab environment
- **Collaboration and Communication**
- Active participation in meetings and team discussions
- Effective sharing of information and coordination between team members
- Responsiveness to feedback and ability to work collaboratively
- **Accountability and Contribution**
- Completion of individual responsibilities as defined in the Team Charter
- Evidence of meaningful contribution to the project
- Ownership of assigned tasks and deliverables

Methodology and Data Sources for Team Performance Evaluation

Team performance will be evaluated through a structured methodology that uses multiple data sources to ensure that assessment is objective, transparent, and evidence-based.

The following methods and data sources will be used:

- **Meeting Records and Participation Tracking**
- Weekly team meetings will be documented
- Attendance, participation, and assigned actions will be recorded
- Meeting notes will provide evidence of engagement and collaboration
- **Deliverable Review and Quality Assessment**
- All project outputs (playbooks, documentation, reports) will be reviewed by the Team Lead and peers
- Feedback and revisions will be tracked to ensure improvement and consistency
- Final deliverables will be assessed against project requirements
- **Execution Observation During IRX**
- Observation of each team member's ability to execute their assigned Red Team playbook
- Monitoring of how effectively tools and techniques are used in the lab environment
- Evaluation of whether expected attack behaviours and outcomes are achieved
- **Milestone and Task Tracking**
- Progress will be measured against the Work Breakdown Structure (WBS)
- Completion of key phases (planning, setup, execution, reporting) will be monitored
- Delays or deviations from the plan will be documented
- **Version Control and Document History**
- Contributions will be tracked through SharePoint document version history
- Individual edits and updates will be identifiable and attributable
- Provides evidence of contribution over time
- **Individual Contribution Log**
- Each team member will maintain a record of their tasks and outputs
- Logs will be used to verify individual effort and responsibility fulfilment
- Supports final evaluation of contribution and performance

7. Risk Management Plan

Purpose: This Risk Management Plan identifies, evaluates, and mitigates risks associated with the cyber security exercise involving multiple attack vectors. It follows the Gelos Enterprises ICT Risk Management Policy and applies the same risk rating methodology (likelihood × consequence). The plan ensures that technical, procedural, and coordination related risks are controlled to maintain exercise integrity and learning outcomes

Risk Register

Risk ID	Risk Description	Likelihood	Consequence	Control Strategy	Resources Required
R1	Technical issues with lab: Lab setup, firewall, IDS, SIEM, or VM configurations may fail during preparation or execution.	Medium	Major	Mitigation: Conduct full pre-exercise testing of all lab components, including firewall, IDS, SIEM, and target systems. Assign an Environment Administrator to troubleshoot issues. Contingency: Use Exercise Plan B with PCAP injection if live attacks fail.	Admin time, Dedicated lab environment, Exercise Plan B PCAP files, Backup VM images/software.
R2	Red Team Issues: Red Team playbooks may not be ready on time, attacks are too weak or unrealistic or	Medium	Moderate	Mitigation: Give each team member clear responsibility for their Red Team playbook and allow enough preparation time. Conduct internal peer review of playbooks and provide technical support during execution.	Clear task allocation, Peer review time, Technical support from Environment Admin, Plan B materials.

	sometimes go out of scope			Contingency: Use Plan B traffic injection if attack execution fails.	
R3	Blue Team Issues: Blue Team procedures may be unclear or incomplete, reducing the effectiveness of incident response. Causing confusion during incidents.	Medium	Moderate	Mitigation: Review Blue Team playbooks before the exercise and brief the Blue Team on expected procedures. Control: Observer/Evaluator provides real-time feedback if the team struggles.	Review time by Scenario Designer/Team Lead, Clear IRP documentation, Dedicated Observer/Evaluator.
R4	Over reliance on tools, under use of procedure : Team waits for alerts or dashboards and ignores manual checks defined in the playbooks	Medium	Moderate	Mitigation: In pre-exercise briefing, emphasise that tools support the process, not replace it. During the exercise, the Observer/Evaluator checks whether the Blue Team follows the written playbook (log review, correlation, communication) instead of only watching the SIEM. Control: Debrief includes a specific section on “process vs tools” and updates playbooks where steps were skipped.	Observer/Evaluator, logging/SIEM access, Debrief checklist focused on procedure adherence.
R5	Poor cross-team coordination : Red and Blue Teams work	Medium	Moderate	Mitigation: Define minimum evidence requirements for each scenario (e.g. key log entries, screenshots of alerts, commands used, containment actions). Assign one Evidence Recorder per team to capture and label artefacts	Evidence Recorder role, Shared evidence folder, Naming convention,

	in silos; timings, injects, and notifications are not aligned, making results hard to interpret.			during the exercise. Control: Immediately after each scenario, teams verify that required evidence has been collected and stored centrally.	time at end of each inject for evidence check.
R6	Evidence collection is incomplete : Screenshots, logs, and decisions are not recorded, making it hard to write the After Action Report or prove what happened.	High	Medium	Mitigation: Define minimum evidence requirements for each scenario (e.g. key log entries, screenshots of alerts, commands used, containment actions). Assign one Evidence Recorder per team to capture and label artefacts during the exercise. Control: Immediately after each scenario, teams verify that required evidence has been collected and stored centrally.	Evidence Recorder role, Shared evidence folder, Naming convention, time at end of each inject for evidence check.
R7	Schedule overrun : Scenarios take longer than planned, so some attacks or responses are rushed or skipped.	Medium	Minor	Mitigation: Prioritise 2to3 core scenarios that must be completed (e.g. phishing, ransomware, web attack), and keep extra scenarios as optional if time allows. Use a strict timetable and give 5-minute warnings near the end of each scenario. Control: If time is short, run the last scenario as a tabletop discussion instead of a full technical run.	Clear schedule, Timekeeper role, simplified (tabletop-only) variant of scenarios

2. Project Management Documentation

For each week throughout our project, we conduct weekly meetings and record minutes from each of these meetings. We took note of the key decisions made for what needs to be done before another meeting in the next week and the progress that each team member made in terms of tasks completed. And we follow up with each other regarding what tasks are done or in progress and any issues that they encountered while working on their tasks.

Project Progress Update - Week 10

Date: 22/04/2026

Attendees: Andrew Byrne, Team member, Team member, Team member, Team member

Key Decisions/Progress:

- Assessment of current progress.
- Multiple members noted they are still finishing up other projects.
- Some members have yet to submit the first part of section 2 that they are assigned to
- Some members have yet to complete the individual part 1.
- Planning Discussion.
- It was discussed and agreed that the overall schedule for project deliverables should be organised by next week. Andrew has agreed to do this.
- Share-point usage
- To assist in organisation and to meet collaboration documentation requirements a versioning system and register will be implemented.
- Andrew has agreed to setup the required files and structure in the delta soc shares folder.
- All files saved to share-point as contribution parts to the assessment shall have the senders' initials as the final part of the file name.

Action Items:

- All team members to have completed assigned tasks for section 2 by next week's meeting
- Andrew to create a schedule for next week's meeting.
- Andrew to organise the share-point folder by next week's meeting.

Next Steps:

- Check in on progress of group deliverables.
- Get feedback on usability of current system for versioning.
- Provide further instruction as required.

Project Progress Update - Week 11

Date: 29/04/2026

Attendees: Andrew Byrne, Team member, Team member, Team member, Team member

Key Decisions/Progress:

- Tasks 1–7 must be completed by Friday. Members who have finished need to copy their work into the original report document don't leave it in Shared files.
- Schedule register has been updated. Members must fill it in for record-keeping purposes.
- The formatting and writing style in the original report document needs to be consistent. The guide document is the bare minimum standard expected.
- Andrew and Team member confirmed they will complete their sections by Friday. The conclusion is assigned to Team member and Team member, which will be completed once Andrew and Team member's part is complete.
- Nessus audit review was discussed to clear up confusion, particularly for Andrew and Team member.
- Referencing style discussed — Harvard style and inline references noted. Any official referencing style is acceptable.
- Quick review of Part 3 conducted.

Action Items:

Action Item	Who	Due Date
Email Steve with team progress update	Andrew	Today
Complete assigned sections and copy to report doc	Andrew, Team member	Friday
Complete conclusion	Team member, Team member	Next week
Tasks 1–7 completed and entered in schedule register	Consult Register	Friday
Prepare 5 ideas for incident response playbook exercise	All	Next Wednesday

Next Steps:

- Next Wednesday Topic: Incident Response Plan (5-point playbook).
- Each member to come up with 5 ideas for exercises to test.
- Playbooks examples: Reconnaissance / Excessive Scanning.

Project Progress Update - Week 12

Date: 06/05/2026

Attendees: Andrew Byrne, Team member, Team member, Team member, Team member

Key Decisions/Progress:

- Reviewed that Part 2 is done and completed and updated in the Register, which is confirmed by Andrew
- Andrew emailed Steve about our progress and our teamwork and cohesion.
- Schedule register has been updated. Members must fill it in for record-keeping purposes.
- The formatting and writing style in the original report document needs to be consistent. The guide document is the bare minimum standard expected.
- Team talked about part 2 in terms of their part in that section and any roadblocks encountered and what to look forward to for part 3.
- Nessus section added to report, confirmed by Team member.
- Discussed 5 ideas from each team for Red Playbooks and what virtual machine to use for these which will be the healthcare VM.
- Decided on the attack chain (i.e. 5 attacks) for the Blue Playbook
 - Reconnaissance - Team member
 - SQL injection - Team member
 - IDOR - Team member
 - REVERSE Shell - Team member
 - PRIV ESC - Andrew

Action Items:

Action Item	Who	Due Date
Complete assigned sections and copy to report doc	Everyone	Next Wednesday
Read through part 3 and its requirements for the cyber project	Everyone	Friday
Have a look at examples of Blue Playbooks templates, etc.	Everyone	Friday
Set up files and other requirements for Part 3	Andrew	Friday

Next Steps:

- Discuss the Blue Playbook examples that we found from today's meeting
- Discuss part 3 of the project in terms of what's required and who is doing which task(s).

Project Progress Update - Week 13

Date: 13/05/2026

Attendees: Andrew Byrne, Team member, Team member, Team member, Team member

Key Decisions/Progress:

- Review of last week meeting points.
- Discussed on the current status of Blue book.
- Discussed on Ransom ware and SQL injection.
- Discussed and concluded that all team members should submit documents in Calibri 12(body) font for project book

Action Items:

- Part-3 should be completed by next Wednesday (20/05/26)
- Generic Pamphlet for team members will be updated by Andrew.

Next Steps:

- Discussion on starting part 4 of the Cyber Security Project

Project Progress Update - Week 14

Date: 20/05/2026

Attendees: Andrew Byrne, Team member, Team member, Team member, Team member,

Key Decisions/Progress:

- Discuss the completion of Part 3 and blue team playbooks
- Team member Team member and Team member will run against healthcare. Two additional Ubuntu machines that Team member and Andrew will use
- Pcap backup plan - Used if something goes wrong - download a pcap file from (web?)trafficanalysis.net
- Part 4 folder has been populated
- Andrew has prefilled parts of the part 4 doc
- Sections are generated and may have inaccuracies that need to be corrected
- Work for the rest of part 4 is evenly distributed
- Red team playbook outlines the attack chain
- Incident response exercise starts next week
- Next meeting is scheduled for 29th of May at 11:00am

Action Items:

- Everyone must produce a red team playbook that goes against their blue team playbook for the same attack
- Everyone needs to have looked at part 5 by Friday

Next Steps:

- Next meeting is scheduled for 29th of May at 11:00am.

Project Progress Update - Week 15

Date: 27/05/2026

No meeting required for that week because we were conducting our IRx at that time.

Project Progress Update - Week 16

Date: 02/06/2026

Attendees: Team member, Team member, Andrew Byrne

Absent: Team member, Team member

Key Decisions/Progress:

- Make sure that all tasks completed are recorded in the register document
- Team member to double check what else needs to be done for part 5 for the IDOR attack
- Team member discussed about his attack in terms of what happened. He needs to do the attack again to get screenshots for his part in part 5.
- Andrew discussed what screenshots to get for our attacks in Part 4, what needs to be done in remaining sections of our project and checked what other team members have.
- Andrew to do part 1 Updated Project Plan, Team member to do part 2 Project Management Documentation and Team member to do Implementation / Work Plan Schedule
- Gather all of our attacks evidence by end of this Friday
- Make sure that our contribution logs and sections in the project documents are done.

Action Items:

Action Item	Who	Due Date
Gather all of our attacks evidence by end of this Friday	Everyone	This Friday
Andrew to do the Updated Project Plan section	Andrew	By next week's meeting
Team member to do the Project Management Documentation section	Team member	By next week's meeting
Team member to do Implementation / Work Plan Schedule	Team member	By next week's meeting

Make sure that our contribution logs are updated	Everyone	By next week's meeting
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Next Steps:

- Check that all of our attacks' evidence are collected
- Check that the Updated Project Plan section, the Project Management Documentation and Implementation / Work Plan Schedule sections are done.
- Follow up with each team members about any pending tasks for the project.

Project Progress Update - Week 17 - 1st Meeting

Date: 09/06/2026

Attendees: Team member, Team member, Andrew Byrne, Team member

Absent: Team member

Key Decisions/Progress:

- There are still parts of the insert document to be completed
- Document management- keep single document on OneDrive and multiple versioned documents locally

Action Items:

Action Item	Who	Due Date
Team member to finish range activities and request a thursday timeslot on the cyber range for his attack.	Team member	This Thursday
Copy the risk register and add a notes page	Team member	By next week's meeting
Team member is to fix the part 5 tables and post in chat it is ready and everyone is to then go and fill in these sections.	Everyone	By next week's meeting
Team member and Team member to complete the evidence file for their section.	Team member and Team member	By next week's meeting
Complete final individual contribution statements.	Everyone	By next week's meeting
Insert completed contingency plan	Team member	By next week's meeting
Complete Purple team activities	Everyone	By next week's meeting
For the Blue team observation chart we shall include the charts	Everyone	By next week's meeting

at the top of our evidence document and include a place holder section in the assessment document pointing to the correct document as example below.		
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Next Steps:

- Follow up with each team members about any pending tasks for the project.

Project Progress Update - Week 17 – 2nd Meeting

Date: 12/06/2026

Attendees: Team member, Team member, Andrew Byrne, Team member

Absent: Team member

Key Decisions/Progress:

- Part 4 insert document is ready.

Action Items:

- Team member and Team member need to put entries into the update register
- Get Team member through the range today
- Team member and Team member need to do Purple teaming activities plan
- Team member and Team member need to fill out their observation checklists
- Andrew has to do section 4 parts 1- 3
- Team member Team member and Team member need to do their final reflections in section 5
- Section 3 part A needs to be completed by Team member Team member and Team member
- Section 4 part 4 – 8 to be completed by Team member and Team member
- Ensure that evidence documents and red books are inserted into their correct folders

Next Steps:

- Each team member to finish off any pending items for the project before finalising all project documents next Tuesday for submission next Wednesday.

Project Progress Update - Week 18 – Final Meeting

Date: 16/06/2026

Attendees: Team member, Team member, Andrew Byrne, Team member, Team member

Key Decisions/Progress:

1. Red Team / Purple Team Activities – Team member

Team member was required to complete Section 3, a (Red Team Activity Playbook) and the Purple Team Activities section, which was initially noted as missing from his submission.

It was later confirmed that Team member had inserted the content into the wrong document. The correct main document has since been updated and finalised with all required content.

2. Part 4 / Part 8 Conclusion – Incident Response Exercise (IRX) – Andrew

The Conclusion section for the Incident Response Exercise (AAR) was assigned to Andrew Part 8

The original document has been completed and updated.

Part 5 – Evidence from IRX Execution: the folder path has been inserted into the original document by Andrew.

3. Appendices – Andrew

A detailed list of documents for the Appendices section will be inserted by Andrew.

4. Final Team Review

Final views were gathered from all team members. All members expressed satisfaction with the team leader and with each other's contributions.

5. Final Submission – ZIP Folder

All project files are to be placed in a ZIP folder inside the SUBMITABLE folder. All content included in the ZIP file is ready for final project submission. Part Included in this folder from 2 to 5. Part is individual work.

Action Items:

- Team member – Confirm Red Team Activity Playbook (Section 3, a) and Purple Team Activities are correctly inserted in the main document.
- Status: RESOLVED – main document updated and finalised.
- Andrew – Complete Part 8 Conclusion for the IRX, inserted Part 5 evidence folder path, and add Appendices document list.
- Status: COMPLETED.
- All Team Members – Ensure all files are included in the ZIP folder inside the SUBMITABLE folder for final project submission.

Next Steps:

- Not Applicable

Risk Management Log

Risk ID	Risk Description	Likelihood	Consequence	Control Strategy	Resources Required
R1	Technical issues with lab: Lab setup, firewall, IDS, SIEM, or VM configurations may fail during preparation or execution.	Medium	Major	Mitigation: Conduct full pre-exercise testing of all lab components, including firewall, IDS, SIEM, and target systems. Assign an Environment Administrator to troubleshoot issues. Contingency: Use Exercise Plan B with PCAP injection if live attacks fail.	Admin time, Dedicated lab environment, Exercise Plan B PCAP files, Backup VM images/software.
R2	Red Team Issues: Red Team playbooks may not be ready on time, attacks are too weak or unrealistic or sometimes go out of scope	Medium	Moderate	Mitigation: Give each team member clear responsibility for their Red Team playbook and allow enough preparation time. Conduct internal peer review of playbooks and provide technical support during execution. Contingency: Use Plan B traffic injection if attack execution fails.	Clear task allocation, Peer review time, Technical support from Environment Admin, Plan B materials.
R3	Blue Team Issues: Blue Team procedures may be unclear or incomplete, reducing the effectiveness	Medium	Moderate	Mitigation: Review Blue Team playbooks before the exercise and brief the Blue Team on expected procedures.	Review time by Scenario Designer/Team Lead, Clear IRP documentation,

	of incident response. Causing confusion during incidents.			Control: Observer/Evaluator provides real-time feedback if the team struggles.	Dedicated Observer/Evaluator.
R4	Over reliance on tools, under use of procedure: Team waits for alerts or dashboards and ignores manual checks defined in the playbooks	Medium	Moderate	Mitigation: In pre-exercise briefing, emphasise that tools support the process, not replace it. During the exercise, the Observer/Evaluator checks whether the Blue Team follows the written playbook (log review, correlation, communication) instead of only watching the SIEM. Control: Debrief includes a specific section on “process vs tools” and updates playbooks where steps were skipped.	Observer/Evaluator, logging/SIEM access, Debrief checklist focused on procedure adherence.
R5	Poor cross-team coordination: Red and Blue Teams work in silos; timings, injects, and notifications are not aligned, making results hard to interpret.	Medium	Moderate	Mitigation: Define minimum evidence requirements for each scenario (e.g. key log entries, screenshots of alerts, commands used, containment actions). Assign one Evidence Recorder per team to capture and label artefacts during the exercise. Control: Immediately after each scenario, teams verify that required evidence has been collected and stored centrally.	Evidence Recorder role, Shared evidence folder, Naming convention, time at end of each inject for evidence check.
R6	Evidence collection	High	Medium	Mitigation: Define minimum evidence	Evidence Recorder role,

	is incomplete: Screenshots, logs, and decisions are not recorded, making it hard to write the After Action Report or prove what happened.			requirements for each scenario (e.g. key log entries, screenshots of alerts, commands used, containment actions). Assign one Evidence Recorder per team to capture and label artefacts during the exercise. Control: Immediately after each scenario, teams verify that required evidence has been collected and stored centrally.	Shared evidence folder, Naming convention, time at end of each inject for evidence check.
R7	Schedule overrun: Scenarios take longer than planned, so some attacks or responses are rushed or skipped.	Medium	Minor	Mitigation: Prioritise 2to3 core scenarios that must be completed (e.g. phishing, ransomware, web attack) and keep extra scenarios as optional if time allows. Use a strict timetable and give 5-minute warnings near the end of each scenario. Control: If time is short, run the last scenario as a tabletop discussion instead of a full technical run.	Clear schedule, Timekeeper role, simplified (tabletop-only) variant of scenarios

3. Completed IRX Deliverables

a. Red Team Activity Playbooks (Consolidated Document)

Summary: This document contains 5 Red Team playbook Summaries developed by team members for the following attack scenarios:

1. REDACTED — peer-authored playbook is not included in the public portfolio version.

PRIVILEGE ESCALATION

Objective: Simulated Privilege Escalation on Debian/UbuntuX System

Test whether the Blue Team can detect and respond to a simulated privilege escalation where an attacker moves from a standard user account to root-level access without using normal administrative methods such as sudo or su.

Prerequisites:

Kali Linux attacker machine, UbuntuX target machine, SSH access to the target, privilege escalation proof-of-concept file, Python HTTP server, wget, Elastic Agent, Security Onion logging, and baseline system logs.

Steps:

1. Preparation:

Confirm the Kali and UbuntuX IP addresses, place the privilege escalation proof-of-concept file on the Kali machine, start a basic HTTP server, confirm logging is active, and verify that baseline activity is being recorded.

2. Execution:

The attacker connects to the UbuntuX target using SSH, runs basic identity-checking commands, downloads the payload from the Kali machine to \$HOME, makes it executable, runs the payload, and then checks whether root-level access has been achieved.

3. Observation:

The Blue Team should monitor for suspicious activity such as external SSH access, payload download from the attacker machine, abnormal privilege escalation behaviour, and root command execution from a non-root starting point.

4. Evidence Collection:

Record timestamps, commands executed, system responses, Security Onion or SIEM alerts, relevant log entries, and screenshots showing detection and investigation evidence.

Expected Blue Team Response:

If the Incident Response Plan is working correctly, the Blue Team should identify the suspicious SSH login, detect the file download and execution, investigate the privilege escalation indicators, classify the incident, collect evidence, and begin containment or escalation procedures.

Success Criteria:

The test is successful if the activity generates visible logs and alerts, the Blue Team can trace the attack chain from SSH access through to privilege escalation, and evidence is collected to support the incident report.

Summary / Excerpt:

This playbook simulates a post-compromise privilege escalation attack against an UbuntuX host. The Red Team uses a Kali system to access the target, transfer and execute a privilege escalation proof-of-concept, and verify root access. The purpose is not only to test the attack path, but to confirm whether monitoring, alerting, investigation, and response processes can detect the full chain of activity.

REDACTED — peer-authored playbook is not included in the public portfolio version.

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b. Exercise Environment Configuration Documentation**Network Topology**

The environment is a Proxmox-based cyber range separated into SOC, Attack, and Target network zones. External access is via a Cloudflare Tunnel, then through an external firewall/vmbr0 path to Router 1 and vmbr1. From vmbr1, RDP access is shown into the SOC workstation area and the Kali attack area.

Router 2 appears to provide internal routing between the SOC, Attack, and Target areas. The SIEM has a management connection into the SOC network and a separate monitoring connection labelled **Monitor (No IP)** toward the target network.

Subnets

- **SOC subnet:** 192.168.100.0/24
- **Attack subnet:** 172.16.150.0/24
- **Target subnet:** 10.2.2.0/24

- **Monitor network:** No IP address assigned

VMs / Systems

System	Role	Network/Subnet
Cloudflare Tunnel	External access tunnel	External
Firewall	External firewall/access control	Connected to vmbr0
Router 1	External/access router	Between firewall and vmbr1
Router 2	Internal router between lab networks	SOC / Attack / Target
SIEM	Security monitoring system	SOC 192.168.100.0/24 + Monitor No IP
Analyst-1	SOC workstation	SOC 192.168.100.0/24
Analyst-2	SOC workstation	SOC 192.168.100.0/24
Analyst-3	SOC workstation	SOC 192.168.100.0/24
Kali	Attacker VM	Attack 172.16.150.0/24
Kali-1	Attacker VM	Attack 172.16.150.0/24
Kali-2	Attacker VM	Attack 172.16.150.0/24
Kali-3	Attacker VM	Attack 172.16.150.0/24
Windows Server	Target VM	Target 10.2.2.0/24
Healthcare	Target VM	Target 10.2.2.0/24
Jangow	Target VM	Target 10.2.2.0/24

Network Connections

- Cloudflare Tunnel connects into the external access path.
- The external access path passes through a firewall and vmbr0.
- Router 1 connects the external/vmbr0 side to vmbr1.
- vmbr1 provides RDP access paths into the SOC workstation area and Kali attack area.
- The SOC network uses vmbr2 and the 192.168.100.0/24 subnet.
- The Attack network uses vmbr3 and the 172.16.150.0/24 subnet.
- The Target network uses vmbr7 and the 10.2.2.0/24 subnet.
- Router 2 connects the internal SOC, Attack, and Target areas.
- The SIEM has a management connection to the SOC network.
- The SIEM also has a monitoring connection labelled **Monitor (No IP)** toward the target network.

System Configurations

Attacker VM

- **Operating System:** Kali Linux
- **Network:** Attack subnet, 172.16.150.0/24
- **Role:** Used by the Red Team to conduct attacks against the target systems.

Victim VM(s)

- **Windows Server**

- Healthcare
- Jangow
- Ubuntu LTS 24

Security VM

- **System:** Security Onion
- **Network:** SOC subnet, 192.168.100.0/24
- **Management connection:** Connected through the SOC management network/vmbr2
- **Monitoring connection:** Separate monitor link labelled **Monitor (No IP)**

Security Tools Configuration

Firewall

A firewall is shown between the Cloudflare Tunnel/external access path and Router 1. It controls traffic entering the Proxmox environment through vmbr0.

SIEM

- **Tool used:** Security Onion
- **Network:** 192.168.100.0/24
- **Management connection:** Connected to the SOC management network
- **Monitoring connection:** Monitor link with no IP address
- **Log sources:** Elastic Agents
- **Alerts configured:** Suricata Default rule sets

Challenges Encountered

- Some issues around bandwidth load and multiple simultaneous logins were experienced during the IRx.

C. Exercise Plan B (Contingency)

Purpose of Plan B

Plan B ensures the cyber exercise continues smoothly if technical issues prevent the live Red Team attack from running as intended.

By switching to pre-prepared PCAP files, the exercise still achieves its learning objectives, maintains scenario integrity, and allows both teams to complete their required tasks.

Plan B Procedure — Actions to Take if the Main Operation Fails

If the live attack cannot proceed due to issues such as VM failure, tool malfunction, or network instability, the exercise transitions to Plan B.

Under Plan B, recorded network traffic (PCAP files) is used instead of live attack traffic.

1. Declare a Controlled Switch to Plan B

- The Exercise Controller formally announces that the live attack cannot continue.
- All participants are informed that the scenario will proceed using recorded traffic.
- The switch is documented for transparency but does not affect assessment.

2. Select the Appropriate PCAP File

- Choose a PCAP file from the prepared Plan B library (sourced from malware-traffic-analysis.net).
- **Select a file that matches the intended scenario type, such as:**
 - o phishing
 - o malware infection
 - o command-and-control (C2) traffic
 - o lateral movement
- This ensures that the exercise remains aligned with the original objectives.

3. Load the PCAP into the Analysis Environment

- The Environment Administrator loads the PCAP into Wireshark or the SIEM replay tool.
- Timestamps and packet order must be preserved to maintain realism.
- Confirm that all participants can access the replayed traffic.

4. Brief the Blue Team

- Inform the Blue Team that they will now analyse replayed traffic instead of live activity.
- No hints or clues about the attack type are provided.
- The scenario continues exactly as if the attack was happening in real time.

5. Blue Team Performs Full Incident Response

The Blue Team proceeds with the same workflow used for live attacks:

- Detect suspicious traffic patterns
- Extract Indicators of Compromise (IOCs)
- Build a timeline of attacker activity
- Identify affected systems or users
- Recommend containment and remediation actions

This ensures the exercise still tests detection, analysis, and response skills.

6. Collect Evidence as Normal

- Screenshots, logs, IOCs, and analysis notes must still be captured.
- Evidence is stored in the shared evidence folder using the standard naming convention.
- This ensures that the After Action Report (AAR) remains complete and accurate.

7. Proceed to Debrief

The debrief covers:

- What the Blue Team detected
- How they analysed the PCAP
- What conclusions they reached
- What actions they would take in a real environment

The use of Plan B is noted but does not reduce the value or validity of the exercise.

D. Purple Teaming Activities Plan & Observation Checklist

Purple Teaming Activities for the IDOR attack:

- The red team launches the IDOR attack on the Electronic Medical Records (EMR) Web Application by changing the value of GET query parameters fname and lname from a url when a patient's details are shown in the healthcare system after logging into the application under the Medical user account. The Blue team logged into Security Onion to confirm that the IDOR attack was detected.
- The red team launches the IDOR attack on the EMR Web application by changing the value of the Issue GET parameter value from 1 to 2 to access another issue of the patient in detail. This attack is repeated by the Red Team from 2 to 3 and from 3 to 4 for the Issue parameter value. The Blue team logged into Security Onion to confirm that the IDOR attack was detected.
- o The Blue team then temporarily restrict access of the Medical user since this attack was done multiple times and within a very short time frame.
- o A ticket was created by the Blue team in Security Onion and sent to the IT Security and Web teams so that they can have a look into this issue further.

Purple Teaming Activities for the Ransomware attack:

- The Red Team used swaks on Kali Linux (172.16.159.111) to deliver a phishing email containing SecurityComplianceTool.py to the victim VM (xUbuntu, 10.2.2.111) via MailHog on port 1025. The victim (Peter Jones) opened and executed the attachment, simulating ransomware installation.
- The Blue Team monitored Security Onion (SIEM) during execution. Zero SMTP alerts were generated due to Zeek monitoring the management interface instead of the attack network interface — a misconfiguration identified post-exercise.

- Post-exercise debrief: Red Team and Blue Team jointly reviewed the SIEM failure. The teacher (Steve) confirmed successful email delivery via direct observation, validating the attack path even in the absence of automated detection.
- Key learning shared across teams: SIEM interface assignment must be verified before any IRX. Both teams agreed that a pre-exercise SIEM connectivity check should be added to the standard IRX checklist for future exercises.

Purple Teaming Activities for the Privilege Escalation attack:

- For the Privilege Escalation exercise, the purple teaming activity focused on sharing knowledge between the Red Team and Blue Team so both sides could understand how the attack worked, how it appeared in Security Onion, and where the detection gaps existed. Instead of treating the exercise as separate Red Team and Blue Team activities, we used the attack as a learning opportunity to compare attacker intent with defender visibility.
- Before reviewing the results, the Red Team provided a short walkthrough of the attack path, including the purpose of each stage: initial SSH access, user enumeration, exploit download, execution from /tmp, and root-level access. This helped the Blue Team understand what behaviours to search for and what evidence should exist if monitoring was working correctly.
- The Blue Team then shared what was actually visible in Security Onion and endpoint telemetry. This allowed both teams to compare the expected detection points against the evidence that was captured. The ELF download alert was used as the main pivot point, while SSH activity, enumeration, and root shell execution were reviewed as examples of activity that may be logged but not always alerted on.
- A joint discussion was held to identify why some stages were easier to detect than others. This helped transfer knowledge in both directions: the Red Team gained a better understanding of which attacker behaviours created useful defender evidence, while the Blue Team gained a clearer understanding of how privilege escalation activity develops from the attacker's point of view.
- The outcome of the purple teaming activity was a set of agreed improvements for future exercises, including improved endpoint process logging, prepared Security Onion Hunt queries, alerts for suspicious execution from /tmp, detection logic for root processes linked to non-privileged user sessions, and a shared timestamp run sheet to make Red Team actions easier to correlate with Blue Team evidence.

Purple Teaming Activities for the SQL Injection (SQLi) attack

- The purple teaming activity for the SQL Injection attack focused on aligning the Red Team's attack behaviour with the Blue Team's detection, analysis, and response capabilities in Security Onion. Both teams worked collaboratively to understand how

each SQLi stage appeared in logs, alerts, and packet level telemetry, and to identify where detection gaps existed.

- Before reviewing the results, the Red Team provided a walkthrough of the full SQLi attack chain, including reconnaissance, login bypass, enumeration, automated exploitation, and credential extraction. This helped the Blue Team understand the intent behind each payload, such as ' OR '1'='1' --, blind SQLi using SUBSTRING(), time delay SQLi using SLEEP(), and UNION based SQLi attempts. These explanations allowed the Blue Team to map each attacker action to expected Suricata signatures and HTTP artefacts.
- The Blue Team then reviewed what was actually visible in Security Onion during the attack. Suricata generated multiple high-severity alerts, including ET WEB_SERVER SQL Injection Select Sleep Time Delay, ET WEB_SERVER SELECT SUBSTRING URI Possible Blind SQL Injection, and ET WEB_SERVER Possible SQL Injection Attempt UNION SELECT in HTTP URI. The Blue Team also identified automated exploitation behaviour through the python requests user agent. This comparison allowed both teams to validate which SQLi stages were detected immediately and which required deeper log analysis.
- A joint discussion was held to compare expected vs. actual detection points. The teams noted that login bypass and enumeration were detected quickly, while some repeated SQLi payloads required correlation across multiple events. The Blue Team explained how they pivoted from Suricata alerts into Hunt queries to confirm exploitation attempts, while the Red Team clarified how automated tools such as openemr410_blind_sql.py generated specific patterns in network traffic.
- The purple teaming session concluded with agreed improvements for future exercises, including refining SQLi related alert thresholds, adding correlation rules for repeated SQLi payloads, improving visibility into automated exploitation behaviour, preparing pre-built Hunt queries for SQLi detection. Both teams also agreed to enhance containment playbooks by integrating faster endpoint restriction and automated IP blocking workflows.

Purple Teaming Activities for the Recon attack:

- The Red Team conducted reconnaissance activities from the attack machine against the Linux-based target system (10.50.30.20). Activities included host discovery, service enumeration, directory enumeration, and authentication observation using tools such as Nmap, curl, wget, and SSH.
- The Blue Team monitored Security Onion throughout the exercise. Security Onion generated alerts and network events associated with port scanning, HTTP requests to sensitive directories (/admin, /backup, and /config), and SSH connection attempts using invalid usernames.

- During the exercise, the Blue Team reviewed Security Onion alerts, packet captures, and event logs to identify the source of the reconnaissance activity and determine which services and resources were being targeted by the attacker.
- Post-exercise debrief: The Red Team and Blue Team jointly reviewed the attack timeline, Security Onion alerts, and collected evidence. Both teams compared the executed reconnaissance activities against the generated detections to identify strengths and gaps in monitoring coverage.
- The exercise demonstrated that Security Onion was able to provide visibility into multiple stages of reconnaissance activity, including network scanning, web resource enumeration, and failed authentication attempts. Based on the findings, both teams recommended refining detection rules and expanding monitoring coverage to improve visibility of early-stage attacker behaviour in future exercises.

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Blue Team Observation Checklists:

Blue Team Observation Checklist for IDOR Attack

Kindly see the attachment 'Attachments folder/EVIDENCE DOCUMENTS/ Evidence_Execution_IRX_IDOR_Attack-Team member.docx' in the Attachments folder.

Blue Team Observation Checklist for Privilege Escallation

Kindly see the attachment 'Attachments folder/evidence documents/IRX_Evidence_Priv_Esc_AB.docx' in the Attachments folder

Blue Team Observation Checklist for Ransomware Attack

Kindly see the attachment 'Attachments folder/EVIDENCE DOCUMENTS/Evidence_Execution_IRX_Ransomware_Team member.docx' in the Attachments folder for the Blue Team Observation Checklist, SIEM screenshots, and all execution evidence.

Blue Team Observation Checklist for SQL Injection (SQLi) Attack

Kindly see the file IRx_Evidence_SQLi_Team member.DOCX in evidence docs located in attachment folder

Blue Team Observation Checklist for Recon Attack

Kindly see the attachment 'Attachments folder/EVIDENCE DOCUMENTS/Evidence_Execution_IRX_Recon_Team member.docx' in the Attachments folder.

IRX After-Action Report (AAR)

Gelos: Incident Response Exercise (IRX) After-Action Report

Date: 12/06/2026

Prepared by: DELTA SOC

1. Executive Summary

The Gelos Enterprises Incident Response Exercise (IRX) was conducted between **20/05/2026** and **05/06/2026** to test the effectiveness of the newly developed Incident Response Plan (IRP) and supporting Blue Team playbooks. The exercise used controlled attack simulations to assess whether Gelos could detect, analyse, contain, and document different types of cyber security incidents under realistic conditions. The scenarios tested included Reconnaissance, SQL Injection, Insecure Direct Object Reference (IDOR), Reverse Shell activity, and Privilege Escalation.

Overall, the IRX showed that the IRP provided a useful structure for responding to incidents, especially during the detection and analysis stages. Security Onion and related evidence sources allowed the Blue Team to identify several attack behaviours and document incident activity. However, the exercise also highlighted areas for improvement, including stronger endpoint visibility, clearer alerting for post-compromise activity, better preparation of investigation queries, and improved coordination between Red Team actions and Blue Team evidence collection.

2. Exercise Objectives and Scope

This project will plan, configure, execute, and evaluate a simulated Incident Response Exercise (IRX) designed to test the effectiveness of Gelos Enterprises' Incident Response Plan (IRP) and the Blue Team playbooks developed in Task 3. The exercise will focus specifically on validating incident response capabilities across five common cyber attack scenarios: Reconnaissance (Recon), SQL Injection (SQLi), Insecure Direct Object Reference (IDOR), Reverse Shell activity, and Privilege Escalation. The project will assess whether the organisation's response processes can operate effectively under realistic attack conditions and support rapid containment and recovery.

Specific activities included within scope are scenario design and development, configuration of the testing environment, deployment of security monitoring and logging tools, execution of controlled attack simulations, Blue Team response activities, evidence collection, and post-exercise evaluation. Each scenario will test different phases of the IRP including detection, analysis, containment, eradication, recovery, and post-incident review to identify strengths and improvement opportunities within current response procedures.

3. Exercise Overview

During the IRX, Delta SOC executed five controlled attack scenarios against the Gelos test environment. The attacks were designed to test different parts of the IRP, beginning with initial discovery and access activity and progressing through web application attacks, command execution, and privilege escalation. Reconnaissance activity was successfully detected, while the other attack scenarios provided opportunities to assess how well the Blue Team could identify suspicious activity, investigate alerts, collect evidence, and follow the response process.

A key part of the exercise was that team members changed roles between attacker and defender activities. After attacks were performed, the attacking team members moved into the defender role to triage and investigate the incidents in Security Onion. This allowed the team to compare the known Red Team actions against the available Blue Team evidence, identify where detections were successful, and highlight where activity was logged but did not generate clear alerts. This process supported both incident response testing and purple team knowledge sharing.

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1. PRIVILEGE ESCALATION PLAYBOOK: ANDREW BYRNE

4. Detailed Findings

Strengths Identified:

- Security Onion successfully detected the ELF executable download in near real time, giving the Blue Team a clear starting point for investigation.
- Hunt allowed the Blue Team to correlate multiple evidence sources, including Suricata alerts, endpoint process logs, endpoint network logs, and system authentication logs.
- Endpoint telemetry confirmed the privilege escalation by showing /bin/sh /tmp/x executing as root from a parent bash session in /home/nopriv.
- The investigation was able to reconstruct the attack chain from SSH access, to file download, to root shell activity.

Weaknesses/Areas for Improvement:

- The initial alert identified the executable download, but it did not automatically explain the full attack chain. Manual Hunt investigation was required.
- SSH login and enumeration activity were not the main alert trigger. This could delay detection if the payload download alert was missed.
- Security Onion case management had some issues, and not all relevant logs could be added cleanly to the case.
- More specific alerting is needed for root processes spawned from non-privileged user sessions.

Unexpected Observations:

- SSH connections from unknown sources did not trigger a suricata alert
- The strongest initial detection was the ELF download alert, but this alone may not be enough to trigger alerts normally.
- The root sshd event needed careful interpretation because sshd can normally run as root. The stronger evidence was the /bin/sh /tmp/x process running as root.
- The attack cleanup removed visible files from the target, but Security Onion still retained enough log evidence to reconstruct the attack.
- The case system in SO is not as user friendly as I expected it to be.

5. Analysis of IRP Effectiveness

The IRP was partially effective and provided a useful structure for detection, analysis, containment, and post-incident review. The Blue Team was able to use Security Onion alerts and Hunt to identify the attacker system, victim system, downloaded exploit file, and root shell activity.

The IRP worked well for investigation and evidence collection because it guided the Blue Team to review alerts, logs, affected users, source IP addresses, and suspicious process activity. However, the exercise also showed that the IRP depends heavily on good endpoint

logging. Network alerts alone were not enough to fully prove the privilege escalation. Endpoint process telemetry was required to confirm that root access had been obtained outside normal sudo or su channels.

Overall, the IRP was effective for triage and analysis, but could be improved with clearer instructions for using Hunt, collecting process-chain evidence, and escalating confirmed privilege escalation incidents.

6. Specific Recommendations for Gelos

- Create specific rules for suricata that alert to SSH connections from unknown sources.
- Create a specific Security Onion detection rule for root shell execution from non-privileged user sessions. This would reduce the time needed to identify privilege escalation.
- Improve endpoint monitoring to ensure process command lines, parent processes, user context, and privilege changes are consistently logged.
- Create a standard Hunt checklist for privilege escalation investigations, including searches for SSH access, wget, executable downloads, /tmp execution, root shells, and suspicious parent-child process chains.
- Improve case management procedures so that screenshots, alerts, and Hunt results can be added to the case cleanly during the investigation.

7. Lessons Learned for Future Exercises

- Start triage from the strongest alert, then pivot into Hunt using attacker and victim IP addresses.
- Do not rely only on Suricata alerts. Endpoint process logs are required to prove local privilege escalation.
- Record exact detection and response times during the exercise instead of reconstructing them later.
- Capture screenshots of important log fields immediately, including timestamp, user, process name, command line, source IP, destination IP, and parent process.
- For the next IRx, prepare saved Hunt queries before the exercise so the Blue Team can investigate faster.

8. Conclusions

The Privilege Escalation IRX showed that Gelos's Incident Response Plan provided a useful structure for detecting, investigating, and documenting a post-compromise attack, but also highlighted important gaps in alerting and endpoint visibility. Security Onion successfully detected the ELF file download and allowed the Blue Team to pivot through Hunt to reconstruct the wider attack chain, including SSH access, exploit execution, and root shell activity. However, several key stages were logged rather than directly alerted, meaning the investigation depended on manual analysis and endpoint telemetry. Overall, the exercise

confirmed that the IRP is effective as a triage and investigation framework, but should be improved with stronger detection rules, clearer Hunt procedures, better case management processes, and more detailed endpoint logging for future privilege escalation incidents.

REDACTED — peer-authored playbook is not included in the public portfolio version.

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4. Evidence from IRX Execution (Compiled)

Please see the attached '/Attachments folder/EVIDENCE DOCUMENTS/' folder for the individual evidence documents for each attack.

5. Final Individual Contribution Statements

Team member - Final Reflection

Role: Scenario Designer

Key Contributions to Implementation:

- Planned, executed, evaluated and reported detection phases for the IDOR attack where I captured screenshots as evidence for these phases and recorded the attacks' details in the Blue Team observation checklist.
- Collected and consolidated all evidence captured during the exercise and analyse the exercise results against the IRx objectives for the IDOR attack.
- Recruited and defined the proposed members of Gelos's Incident Response Team, defined their specific roles and responsibilities within the IRT (e.g., Incident Commander, Forensic Analyst, Communications Lead, Legal Counsel representative) and outlined the essential training requirements necessary for each role to ensure they can effectively perform their incident response duties.

Challenges Overcome:

- Since I wasn't familiar with how the IDOR attack can be detected and analysed, I run a few test runs of that attack on my virtual machines so that I can be familiar with the detection and analysis of the IDOR attack using Security Onion.
- I was sometimes slow in taking minutes in team meetings so I was able to improve the efficiency and accuracy of taking minutes by taking summarised notes and further elaborate these notes into detailed minutes so that every team member can understand

Skills Developed:

- Creating Blue and Red Team playbooks for a vulnerability / attack such as the IDOR attack, and launching attacks and take action after vulnerability / attack detection based on these playbooks.
- Taking minutes during team meetings and improve my team collaboration with other team members.
- Using Security Onion to detect and analyse malicious attacks and create / escalate cases for these attacks.

Would Do Differently:

- Practice taking minutes by conducting dummy team meetings so that I can improve my speed in taking minutes.
- Be more familiar with the Security Onion platform in detecting and analysing vulnerabilities and attacks so that I can spread my knowledge and expertise to my other team members.
- Co-ordinate with other team members to make sure that detected attacks / vulnerabilities are escalated to relevant departments for further investigation in quick response time.

Andrew Byrne - Final Reflection

Role: Team Leader / Privilege Escalation Scenario Contributor

Key Contributions to Implementation:

- Acted as Team Leader for the Datatrust Gelos project.
- Organised team activities and communications including the documentation structure and contribution tracking.
- Executed the Privilege Escalation playbook during the IRx.
- Performed the Blue team IRx roles and executed the Privilege Escalation Blue team Playbook.
- Analysed the effectiveness of the IRP by identifying which stages were detected by alerts and which were only visible through logs or endpoint evidence. This included noting that the file download generated a clear alert, while SSH login, enumeration, and root shell activity required manual investigation.
- Contributed to the purple teaming activity summary by linking Red Team actions with Blue Team observations and documenting what should be improved in future exercises, including stronger endpoint process logging and privilege escalation alerting.

Challenges Overcome:

Coordinating a small team who were all working remotely and using shared resources provided some logistical challenges and co-ordination. Clear communication was required to facilitate effective group activities and project deliverables.

The privilege escalation attack did not generate clear alerts for every stage, so I had to work backwards from the ELF download alert and use Security Onion Hunt to identify the wider attack chain.

Some activity occurred inside the SSH session, meaning commands such as whoami, id, groups, and sudo -l were not fully visible through network IDS alerts. This helped me understand the difference between network detection and endpoint-level process visibility.

Correlating the Red Team transcript with Security Onion evidence was challenging because the investigation required matching timestamps, attacker IP, victim IP, user account, downloaded file, and process execution evidence.

Skills Developed:

- Using Security Onion Hunt to investigate attacker activity, pivot between related events, and confirm suspicious behaviour across network and endpoint evidence.
- Analysing privilege escalation indicators, including abnormal execution from /tmp, suspicious file download activity, and root shell execution linked to a non-privileged user session.
- Recording incident evidence in a structured way using timestamps, affected hosts, source IPs, detection status, response actions, and IRP stages.
- Understanding the practical limits of SIEM and IDS visibility, especially where activity is logged but does not automatically generate an alert.
- Team leadership and management skills.
- Clearer communication skills.

Would Do Differently:

- Prepare specific Security Onion Hunt queries before the live exercise to be included in the blue team playbook to make the triage process faster and more consistent.
- Record exact attack timestamps in a shared run sheet during execution so that Red Team and Blue Team evidence can be correlated more quickly.
- Create or tune alerts for suspicious execution from /tmp, external file downloads, and root processes started from non-privileged user sessions.
- Conduct more practice investigations in Security Onion before the final exercise so that response time can be reduced and evidence collection can be completed more confidently.

Team member-Final Reflection

Role:Observer/Evaluator

1.Key Contributions to Implementation

I executed the full SQL Injection attack chain, including reconnaissance, login bypass, database enumeration, automated blind SQLi, and credential extraction. I also monitored how each stage appeared in Security Onion and recorded all timestamps, alerts, and evidence for the IRX. My observations helped evaluate how well the Blue Team followed the IRP and whether their SQLi playbook worked in a real-time scenario.

2. Challenges Overcome

Some SQLi stages did not immediately correlate in Security Onion, especially blind SQLi and repeated payloads, so I had to rely on manual Hunt analysis to confirm exploitation. Automated exploitation using python-requests also created patterns that were not originally documented in the playbook. Despite these challenges, I was able to clearly map each attacker action to the correct detection points and gather complete evidence for the AAR.

3. Skills Developed

This exercise strengthened both my offensive and defensive skills. I learned how attackers quickly escalate from a simple input field to full system compromise, and I improved my ability to analyse Suricata alerts, Hunt logs, and SQLi patterns such as SUBSTRING(), SLEEP(), and UNION-based payloads. I also gained confidence in evaluating IRP effectiveness, understanding SIEM visibility, and identifying gaps in detection and containment workflows.

4. What I Would Do Differently

Next time, I would prepare prebuilt Hunt queries for SQLi to reduce manual correlation time. I would also update the SQLi playbook to include automated exploitation indicators, improve alert tuning, and recommend faster containment steps such as automatic IP blocking. Finally, I would validate SIEM interface monitoring before the exercise to ensure no SQLi traffic is missed.

Team member - Final Reflection

Role: Environment Administrator

Key Contributions to Implementation:

Configured the xUbuntu victim VM with MailHog SMTP catcher, verified network connectivity between Kali Linux and xUbuntu, and prepared the lab environment for the Ransomware IRX.

Developed and executed the Red Team Ransomware attack using a phishing email (swaks) to deliver SecurityComplianceTool.py to the victim VM, and documented all execution evidence with screenshots.

Completed IRX AAR sections 4–8 for the Ransomware attack, including an analysis of the SIEM configuration gap, and produced a separate evidence document filed in the Attachments folder.

Challenges Overcome:

Security Onion's Zeek sensor was monitoring the wrong network interface, resulting in zero SIEM detection during the exercise. I escalated this to the team lead immediately and worked with the teacher (Steve) to confirm attack delivery via verbal observation.

The swaks command syntax required adjustment during the exercise as the lab guide's flags differed slightly from the installed version; I adapted the command on the fly to achieve successful email delivery.

Skills Developed:

Configuring and administering a virtual lab environment including MailHog SMTP catcher and verifying network connectivity between Kali Linux and xUbuntu VMs.

Designing and executing a phishing-based ransomware attack using swaks, and documenting technical evidence for both reporting and assessment purposes.

Analysing SIEM tool configuration gaps and understanding the difference between a functioning IRP and the monitoring capability required to detect real-world attacks.

Would Do Differently:

Verify SIEM interface configuration before the exercise begins, confirming Zeek is monitoring the attack network rather than the management interface.

Test all attack tools including swaks in a dry run against the actual lab environment to catch syntax or connectivity issues before the live IRX.

Prepare contingency evidence such as pre-captured PCAP files in advance, so that if live SIEM detection fails, there is still a record of the attack for the Blue Team to analyse.

Team member – Final Reflection

Role: Observer/Evaluator

Key Contributions to Implementation:

- Developed the Reconnaissance Blue Team playbook and Red Team playbook.
- Planned and executed the reconnaissance attack simulation against the Linux-based target system.
- Analysed Security Onion alerts, events, and collected evidence generated during the exercise.
- Created the attack timeline, observation checklist entries, and evidence collection requirements.
- Contributed to the incident response exercise documentation and final reporting.

Challenges Overcome:

- Ensured the reconnaissance activities generated sufficient evidence for Security Onion detection.
- Correlated attacker actions with Security Onion alerts and event logs.
- Identified which alerts and logs were most relevant to the incident response process.
- Maintained alignment between the Red Team attack chain and Blue Team response procedures.

Skills Developed:

- Improved understanding of reconnaissance techniques and attack methodologies.
- Developed practical experience using Security Onion for monitoring and analysis.
- Enhanced skills in log analysis, evidence collection, and incident investigation.
- Improved technical documentation and cyber security reporting skills.
- Gained experience in creating and evaluating incident response playbooks.

Would Do Differently:

- Test a wider range of reconnaissance techniques to generate additional attack data.
- Spend more time refining Security Onion alert configurations before the exercise.
- Collect more screenshots and evidence during the attack simulation.
- Measure detection and response times more accurately throughout the exercise.
- Conduct additional exercise iterations to further evaluate Blue Team readiness.

APPENDICES

Appendix A: Complete Red Team Playbooks

The complete Red Team playbooks are provided as supporting documents in the **Attachments folder/RED BOOKS** directory. These documents contain the full attack plans, objectives, prerequisites, execution steps, and expected detection opportunities for each IRX scenario.

Playbook	File Path
IDOR Attack Red Team Playbook	Attachments folder/RED BOOKS/Team member-Red_Playbook_for_IDOR_attack.docx
Privilege Escalation Red Team Playbook	Attachments folder/RED BOOKS/red_team_playbook_full_priv_esc_AB.docx
Ransomware Red Team Playbook	Attachments folder/RED BOOKS/RedPlayBook_Ransomware_Daud.docx
Reconnaissance Red Team Playbook	Attachments folder/RED BOOKS/RedPlaybook_Recon_Team member.docx
SQL Injection Red Team Playbook	Attachments folder/RED BOOKS/SQLi_RedPlaybook_Team member.DOCX

Appendix B: Complete Evidence Package

The complete evidence package is provided in the **Attachments folder/EVIDENCE DOCUMENTS** directory. These files contain screenshots, observations, logs, and supporting documentation collected during the IRX execution and Blue Team investigation activities.

Evidence Document	File Path
IDOR Attack Evidence Document	Attachments folder/EVIDENCE DOCUMENTS/Evidence_Execution_IRX_IDOR_Attack-Team member.docx
Ransomware Attack Evidence Document	Attachments folder/EVIDENCE DOCUMENTS/Evidence_Execution_IRX_Ransomware_Team member.docx
Reconnaissance Evidence Document	Attachments folder/EVIDENCE DOCUMENTS/Evidence_Execution_IRX_Recon_Team member.docx
Privilege Escalation Evidence Document	Attachments folder/EVIDENCE DOCUMENTS/IRX_Evidence_Priv_Esc_AB.docx
SQL Injection Evidence Document	Attachments folder/EVIDENCE DOCUMENTS/IRx_Evidence_SQLi_Team member.DOCX

Appendix C: Meeting Minutes and Project Communications

Project communication records, meeting minutes, attendance records, agenda items, and supporting templates are provided in the **Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS** directory. These files document team communication, planning decisions, attendance, and project coordination during the IRX planning and implementation phases.

Communication / Project Document	File Path
Agenda Items	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/Agenda_Items.xlsx
Attendance Register	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/attendance_register.xlsx
Meeting Minutes - 02/05/2026 - AB	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_02052026_AB.docx
Meeting Minutes - 06/06/2026 - DT	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_06062026_DT.docx
Meeting Minutes - 02/06/2026 - AM	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_020626_AM.docx
Meeting Minutes - 05/06/2026 - Team member	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_050626_Vamsi.docx
Meeting Minutes - 06/05/2026 - AM	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_06052026_AM.docx

Meeting Minutes - 09/06/2026 - AB	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_09062026_AB.docx
Meeting Minutes - 12/06/2026 - OB	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_12062026_OB.docx
Meeting Minutes - 13/05/2026 - VM	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/Minutes_13052026_VM.docx
Meeting Minutes - 20/05/2026 - OB	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_20052026_OB.docx
Meeting Minutes - 22/04/2026 - AB	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_22042026_AB.docx
Meeting Minutes - 29/04/2026 - DT	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/minutes_29042026_DT.docx
README File	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/README.md
Meeting Minutes Template	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/template.docx
Meeting Minutes Template Backup	Attachments folder/COMMUNICATION EVIDENCE DOCUMENTS/template_backup.docx

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